

Speech systems

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With the new and increasingly sophisticated network services being offered by the intelligent network comes the problem of how the customers control these services. Although new telephones are being developed to assist with this problem, the majority of customers will, for the foreseeable future, have only the telephone keypad/dial and handset as a means of control. As a result, the ability to provide a 'user friendly' method of controlling these services via the telephone will be essential in ensuring their success and ultimately the success of the intelligent network.

Using speech for communication between the customer and the network is seen as fundamental to the operation of the intelligent network. This paper reviews the role of a speech system in the telephone network, and describes how the BT speech applications platform has been designed to meet the technical systems requirements for this role. The paper concludes with an overview of how the BT speech applications platform is being developed to function as an intelligent peripheral within the intelligent network.

1. Introduction

The perceived quality of the services offered by a telecommunications company, and thereby the perceived quality of that company is, in part, derived from experiences which customers have when using or encountering network and voice services. In particular, such perceptions tend to be based upon the ease of use and naturalness of these services. As such, it is important that voice automation and voice services are used in appropriate circumstances and are designed and structured so as to improve perceptions rather than lower them.

Managing customers' expectations is a subject that all companies wrestle with on a day-by-day basis and a comprehensive treatment of this subject can be found in Atyeo and Green [1]. However, a brief review of the human factors aspects of telephone network services is presented here to illustrate some of the problems facing designers of speech services.

By way of an example, consider making a telephone call to a mail order company. In this case, the customer would expect the call to be answered quickly and would wish to be dealt with in a polite and efficient manner. Even if the mail order company in question cannot provide the requisite service, they would wish to leave the customer with a good impression, so facilitating repeat business. Most companies strive for this goal when dealing with customers on the telephone, be it via human or automated operators. However, in practice, it is clear that the customer's expectation of an interaction with a machine is different to

that of interacting with a person. Imposing a badly designed automated service or insisting that callers interact with automated services in inappropriate circumstances can ultimately be damaging to the business. Conversely, appropriate use of well designed and implemented services can significantly enhance customer and caller perceptions.

In many cases, given the choice of interacting with a person or a machine, callers would not choose to interact with an automated service. An illustration of this is the response which many people have when confronted with an answering machine. In some cases, the caller will simply hang-up and not leave a message. As the caller may be disappointed at not having made direct contact with the person they have called, this behaviour is accepted, but is difficult to explain rationally. In addition, the owner of the answering machine has installed the answering machine not only for their own convenience but also for that of the caller.

Although this is a simple example, it demonstrates that customers' expectation and attitude on meeting an automated speech system differs considerably to that of a person-to-person interaction. As such, a key problem facing developers of all automated voice services is how to manage the interaction, such that callers are positively encouraged to continue the dialogue with the speech system, despite a possibly negative attitude to such interactions. This problem is all pervading and at the current time is relatively independent of the specific technology or

architectural approach used to deliver the network or speech service.

2. Speech systems — a functional overview

A speech system in a telephone network has three principal functions:

- to provide general voice guidance, e.g. informing a customer when they have dialled an unrecognisable telephone number,
- to provide customers with automated control of network services, e.g. invoking a call diversion facility,
- to provide voice services, such as call completion, voice messaging, information provision (e.g. Directory Enquiries), etc.

A speech system can be broken down into a number of key components which are shown in Fig 1.

The nature and function of each of these components is discussed below.

2.1 Telephony interface

The telephony interface can be divided into two functional areas — the speech interface and the signalling interface.

In systems with relatively few lines (up to ten, say), the telephony interface will usually be an analogue connection. The signalling available from an analogue interface is limited to a small number of call progress tones, e.g. equipment busy tone, number unobtainable, which the speech system has to automatically detect in order to manage the call.

When the number of lines becomes large it is far more economical to use a digital telephony interface. This brings the added benefit of improved speech quality. Signalling over digital connections comes in two forms, channel associated and common channel. Although channel associated signalling is much quicker and more reliable than analogue signalling, the call management information is essentially the same as for the analogue interface. With this form of signalling, the analogue supervisory tones are allocated digital codes which are then embedded within the speech channels and can be detected and interpreted by the speech system.

However, common channel signalling, where the signalling and speech information are transmitted separately, contains extensive call-related data in addition to basic call progress information. This additional information can be used by a speech system for improving the quality or ease of an interaction with customers and callers. One example of such information is calling line identity (CLI). This can be used by the speech system to identify the customer, prior to any interaction with the speech service taking place, and allow customer specific information to be accessed, thus improving the perceived quality of the

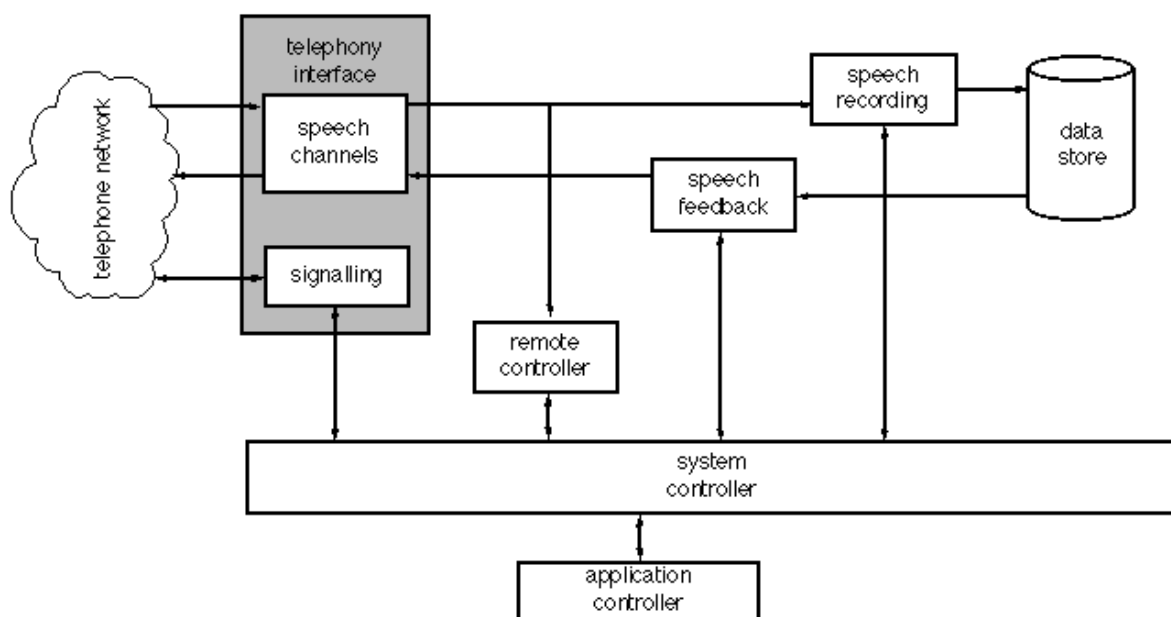


Fig 1 Speech system functional block diagram.

interaction. For example, the speech system might inform the customer which network facilities were currently enabled on their telephone line.

2.2 *The remote controller*

The purpose of the remote controller is to detect the control signals sent from the customer's telephone. The customer has two methods available for remotely controlling a service using the telephone — the keypad or dial, and the microphone, i.e. using speech.

The telephone dial or keypad were designed purely for call set-up and hence signalling from the telephone to the local exchange. With the advent of TouchTone signalling it was quickly realised that these tones could be transmitted across an established telephone connection (after call set-up) and reliably recognised at the receiving end. In the UK, however, there are some telephones that use loop-disconnect signalling, especially in the residential market. As a result, and in order to provide a ubiquitous service, it is necessary to consider the use of loop-disconnect transients¹ as a means of remotely controlling network services; however, the speed of transmission and the accuracy of recognition for loop disconnect transients is lower than that for TouchTone signalling.

In operation, the remote controller recognises the signals sent by the customer's telephone and transmits them to the application controller (see section 2.7). The number of different signals that can be originated by a telephone dial/keypad is limited to twelve — the digits '0 to 9', '*', and '#'² — although loop-disconnect telephones usually transmit the digits only.

An alternative approach to providing callers with the ability to control network services from their telephone is to use voice recognition. In this case, the customer speaks appropriate commands into the telephone microphone. Using speech commands in this way has two principal advantages over the keypad/dial:

- there is, in theory, no limit to the size of vocabulary that is available,
- control of the service can be made to appear more natural and, therefore, easier.

¹ A telephone produces loop-disconnect signals simply by making and breaking the direct current on the local telephone line. During call set-up these direct current pulses are terminated at the local exchange and are not transmitted any further. If these pulses are transmitted once a call is set up, a series of transients is induced on the line beyond the local exchange and these can be detected at the receiving end of the telephone connection.

² There are 16 TouchTone tones; however, only 12 of these are used by most telephones.

To facilitate this type of control, the speech system must be able to recognise the speech commands spoken by the customer. Until recently the quality of speech recognition technology was not perceived as being acceptable for deployment in network services. This perception was reinforced to a degree by an early approach to the use of speech recognition, which was to simply replace TouchTone key presses with the spoken form (i.e. rather than dialling/pressing the digit '1', the customer was invited to say the word 'one'). This particular approach proved to be less than satisfactory for a number of reasons. Not only are the spoken digits notoriously difficult to recognise automatically — they appear to a recogniser to be very similar acoustically — but the control of the service was slower than using the keypad/dial. As a result customers became frustrated with the early voice driven services.

The utilisation of speech recognition as a form of remote control has, however, progressed substantially in recent years through advances both in the performance of the underlying recognition algorithms and the price/performance of the hardware needed to run these computationally intensive processes. As the technology has advanced, the application of speech recognisers has tended to drift away from being simply digit detectors to that of assisting users to invoke facilities in a more natural fashion. To this end speech recognition now tends to be used primarily to identify service command words.

An example of where the speech recogniser can be used in this way is in the customer invocation of network services. Currently, a customer invokes a network service, such as call diversion, by entering a unique code made up of TouchTone digits. However, customers may forget the code for the network service they wish to use. Using a speech recogniser the customer could simply speak the name of the service required, e.g. 'divert', and the speech system would transmit the appropriate service code to the network. The keypad is used only for entering telephone numbers, e.g. destination telephone numbers for diverted calls. If the customer does not have a telephone with TouchTone signalling then the speech recogniser can be used for recognising digits or, alternatively (but with less reliable results), loop-disconnect transient recognition may be used.

In order to optimise the accuracy of the speech recogniser the number of words to be recognised at any one time needs to be kept as small as possible and the words themselves need to be easily distinguishable. As such, allowable key words that are used within a service need to be carefully chosen to ensure optimum performance. In the same way, the prompts or guidance messages given to the customer need to be worded carefully to guide the customer, politely and unobtrusively, to use the appropriate words.

2.3 *Speech feedback*

Speech feedback to the customer can be in the form of supervisory tones, speech (via the telephone loudspeaker), or visual indication (by utilising the display that can be found on some telephones). Supervisory tones are still widely used for feedback to the customer, e.g. dial tone, busy tone. However, speech feedback is increasingly being deployed to improve the customer interface, e.g. the BT telephone network now plays a message to a customer when it doesn't recognise a dialled number instead of playing 'number unobtainable' tone. Visual feedback is increasingly being used as telephone instruments evolve to include display capability; this particular form of interaction is relatively new and is not considered further in this paper.

The quality of the voice used in speech feedback is an important factor in creating a favourable impression with customers. Quality is both subjective and nebulous, however, a few basic attributes can be identified. The intonation of the voice should create an impression of helpfulness and politeness, and yet be assertive when required. For example, it may be necessary at some point in the dialogue to instruct the customer to undertake a simple action. The pace of the voice should be such that the customer does not feel rushed. The type of voice selected can also set the perceived 'style' or 'personality' of the company providing the service. For example, a Scottish company may purposely choose a Scottish accented voice for its automated speech services.

2.4 *Speech recording*

Some applications may necessitate the storage of customer information. However, speech requires large amounts of storage — for example, one minute of speech recorded at the standard digital pulse code modulation (PCM) telephony transmission rate of 64 kbit/s requires 480 kbytes of storage. At this rate the speech storage soon becomes a limiting factor with respect to cost and the time taken to retrieve the data. As a result, most speech systems employ some form of data compression to reduce the amount of speech storage required.

The most common compression rate currently being used in commercial speech systems is 32 kbit/s, i.e. one half of the standard digital telephony transmission rate. Using speech compression also helps reduce the internal data transmission requirements within a speech system, ultimately helping to reduce costs.

2.5 *Data store*

Most applications require the storage of customer information. This information is usually in two forms:

- service information, e.g. what services the customer currently has available — this is sometimes referred to as the customer profile,
- application information, e.g. in a call-answering application a caller may leave a speech message for the customer.

This information is ideally suited to the ordered storage methods offered by standard database structures, e.g. relational databases.

2.6 *The system controller*

The system controller has two principal functions:

- to provide a simple means of managing the functions of the speech system,
- to undertake housekeeping to ensure that the speech system is functioning optimally.

2.7 *The application controller*

The role of the application controller is to co-ordinate the elements of the speech system in order to execute a defined transaction, or dialogue, with the customer/caller. All the functions described above (i.e. speech feedback, speech recognition, TouchTone recognition, etc) can potentially be used for interacting with a customer. The only differences between the dialogues for different applications are:

- the order in which the functions are implemented,
- the feedback messages given,
- how the recognised TouchTone tones/loop-disconnect pulses are interpreted,
- the words to be recognised,
- customer or caller information that is to be recorded and stored.

The applications controller makes requests to the systems controller to perform some task, e.g. record the customer's voice.

¹ The most common data compression technique used is Adaptive Differential Pulse Code Modulation (ADPCM).

Having reviewed the functional aspects of a speech system the remainder of this paper is dedicated to describing BT's latest network-based system, the BT speech applications platform (SAP), and how it has been designed to meet BT's requirements for supporting future telephone network-based voice services.

3. The BT speech applications platform — the design philosophy

The speech system that has been deployed into the BT telephone network for supporting new voice services, such as call minder, is the BT speech applications platform (SAP). The SAP has been designed to function in both telephone network and private system environments. Although the SAP equipment used in these environments is similar, each imposes its own set of operational requirements.

At the outset of the SAP development an underlying design philosophy was adopted to try and ensure that a number of critical success criteria were met. This underlying approach and a description of the system structure is given in the following sections.

3.1 *Scaling*

The principal objective for a telephone network designer is to provide services to the customer at optimal cost. As part of the telephone network, the speech system should also be designed to meet this objective. One significant method of cost optimisation is to share the available resources between as many different applications as possible.

To achieve this objective the SAP was designed to be as modular as possible, not only at the macro level from a functional unit perspective, e.g. application processor, but also at a micro level within each functional unit. This is described more fully below.

3.2 *Modularity*

Any large system such as the SAP can be architecturally divided into smaller functional units. The SAP is split into six processing units:

- telephony,
- signal processor,
- customer management unit (file-store),

- system management,
- system back-up and data collector,
- local area network (LAN).

The function of each of these units is described later in the paper. Each unit functions autonomously, hence the SAP has a distributed processing architecture.

Communication between all units is via the LAN. Additional units can be added to the system to meet service requirements and the system re-configured to optimise cost. If, for example, a new service were to be launched where a SAP installation had spare customer management unit (file-store) capacity, then it would only be necessary to add the appropriate signal processor units required to handle the additional traffic.

The functional units are modular in design and can be configured to suit the service requirements. Each unit has a finite amount of processing capability, and yet the processing requirements for each service differ. In order to match the system to the processing needs of the application, the configuration of each unit can be adjusted to optimise the usage of processing resources and hence optimise the cost of providing the service. This technique of configuring resources to suit the service is known as 'dynamic resource allocation' and is described further in the section on cost optimisation (see section 3.3).

Finally, being autonomous processing units means that development on each can take place independently. For example, the telephony unit can be enhanced to exploit a new signalling protocol without affecting the other units in the system.

3.3 *Cost optimisation*

The ultimate performance goal of any service-providing system is to supply the requisite amount of resource to meet a pre-specified level (quality) of service. Providing too much resource incurs a cost penalty and providing too little means a reduction in the level of service. For example, telephone exchanges are designed to meet a specified grade of service which will dictate the maximum number of calls that will be handled in the busiest hour of the busiest day.

The majority of speech platforms are designed such that there is a fixed amount of speech processing resource assigned to each telephone channel. If the service that runs on the platform does not require all the assigned resource then there is a cost penalty. At the other end of the spectrum, the maximum amount of resource available to a service is limited, which can limit the services that can be supported.

To overcome this limitation the resources on the SAP are treated as a single pool and the requisite amount of resource is allocated to a service as required. During the execution of a service, the amount of resource required will depend on the function being executed. For example, the playing of a speech feedback message requires only a small amount of resource compared with executing a speech recognition function. Using this approach, only the required level of resource need be made available to execute any particular function. The net effect is that the resources are employed to the maximum efficiency and thus the cost of the service is optimised. This technique, dynamic resource allocation, is one of the features that distinguishes the SAP from other speech systems available today.

3.4 Industry and open standards

Industry standard components and open standards are employed on the SAP wherever possible providing the following advantages:

- the market for industry-standard components is very active and very competitive, resulting in a wide range of products being available to system developers — the diversity of the market means that a single source of supply is rarely a problem and that prices are keen,
- companies in the market are continually developing new and improved products with better price/performance,
- the companies in the market will develop new products to meet the latest standards.

The net effect of all the above points is that the system designer has at their disposal a number of cost-effective, high-powered computational building blocks that can be used to build an optimum system from both a cost and performance perspective. Upgrade capabilities are provided regularly through the market-place, giving a straightforward future-proofing route.

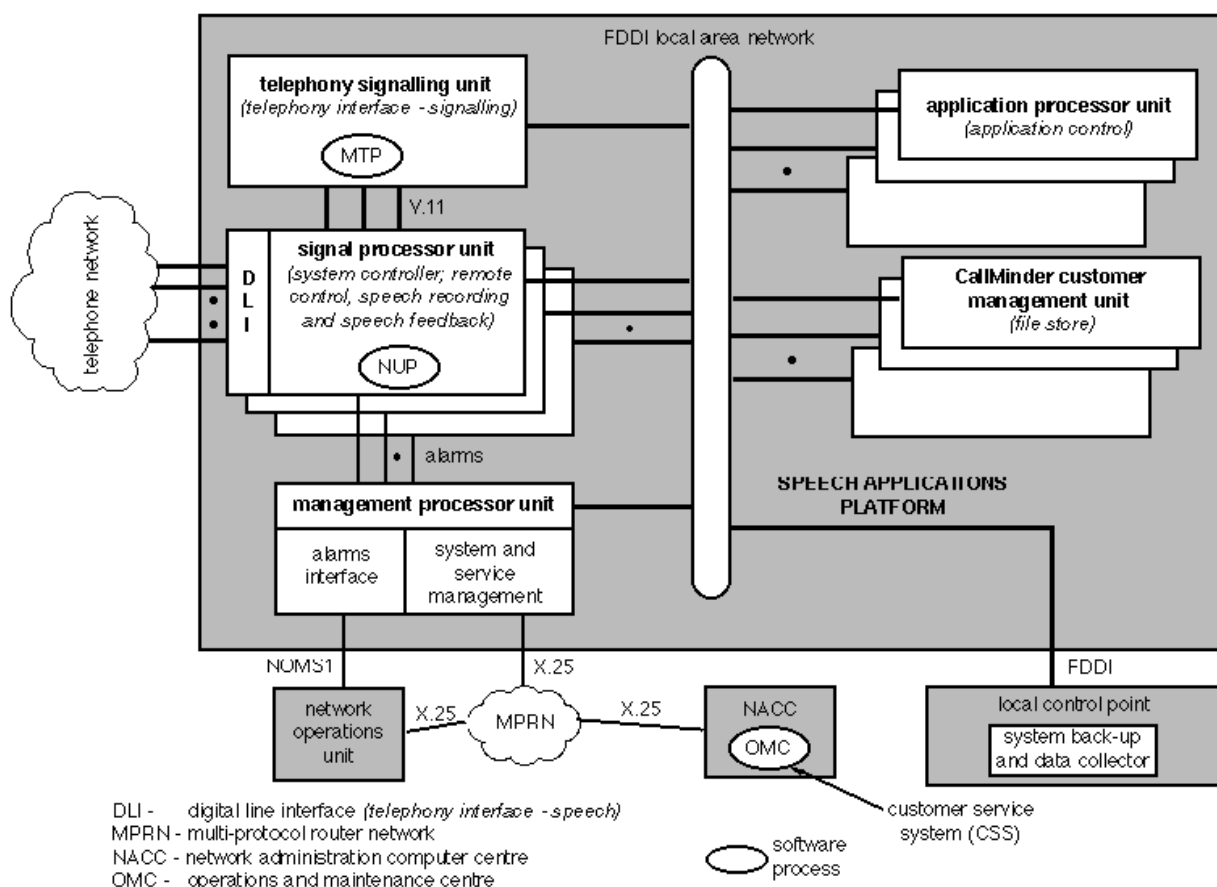


Fig 2 The BT speech applications platform network element.

4. The BT speech application platform — the technical solution

As noted previously, the BT SAP consists of six processing units. Figure 2 shows a functional block diagram of a SAP network element for supporting new voice services such as the BT call minder call answering service in the BT telephone network. The nomenclature in *italics* in Fig 2 corresponds to the functional units described in section 2. This system has been connected into 21 sites in the BT telephone network primarily at digital main switching units (DMSUs). The system is based on the industry standard VME Bus which is used for transmitting control information between the various elements within each processing unit. The function of each processing unit within the SAP is described below.

4.1 *Telephony signalling unit (telephony interface — signalling)*

All telephony communication, both speech and signalling, is provided by standard 30-channel PCM systems, terminated on the digital line interface on the signal processor unit (see section 4.3). Each channel provides a 64 kbit/s bearer which in the case of speech is companded using A-Law encoding (note that the telephony signalling unit handles the signalling functions of the telephony interface).

The signalling channels are separated from the speech channels on the digital line interface. The signalling information is then transmitted to the telephony signalling unit (via a V.11 interface). The signalling employed is ITU-T common channel signalling system number 7 (C7). The telephony signalling unit handles levels 2 and 3¹ — message transfer part (MTP) — of the signalling protocol and extracts the level 4¹ call processing information — network user part (NUP). The NUP contains the call processing functionality required to carry out the network service, e.g. status of line, CLI. The NUP is transmitted to all the signal processor units via the FDDI LAN.

The telephony signalling unit contains three signalling links from separate parent telephone exchanges — this arrangement is sometimes referred to as triple parenting. This is to enable alternative network routing in case of traffic congestion or exchange failure.

¹ These C7 signalling levels do not correspond to ITU-T open systems interconnection (OSI) layers.

4.2 *Application processor (application control)*

The function of the application processor is to run the applications software. As mentioned earlier, each application is a dialogue transaction with the customer/caller.

An application process manages the telephone call via the telephony signalling unit and makes requests to the signal processor unit for signal processing functions via an object oriented application programmers interface (API). The API is a library of high-level functions that allow an application to be developed while at the same time shielding the application developer from the underlying complexity of the SAP system. For example, the application processor could issue the request 'play stored message No X to telephone channel No Y' without any knowledge of the type of telephony interface or the compression rate at which the message was recorded.

The application processor can be either located in each of the signal processing units, or as a separate central resource that communicates with each signal processing unit via the LAN, as is the case in the service node architecture (see section 5). For clarity, Fig 2 shows the latter configuration; however, the SAP systems integrated into the BT's telephone network have the signal processor units and application processors collocated on the same shelves.

4.3 *Signal processing unit.*

The signal processing unit performs the roles of system controller, remote controller, speech feedback, and speech recording. This unit is shown in Fig 3.

As the system controller, this unit is effectively the heart of the SAP and is divided into five main modules.

- Digital line interface (telephony interface) — this unit terminates all digital telephony links and separates the signalling information from the speech data. Most of the data channels carry digitised speech signals and are transmitted to the signal processors via the SpeechBus.
- SpeechBus — it is possible to transmit speech data over the VME control bus; however, as mentioned earlier, digital speech signals have a high data rate and the available bandwidth on the control bus would quickly be exhausted. This would limit the amount of control information that can be transmitted with the result that less voice services can be supported per

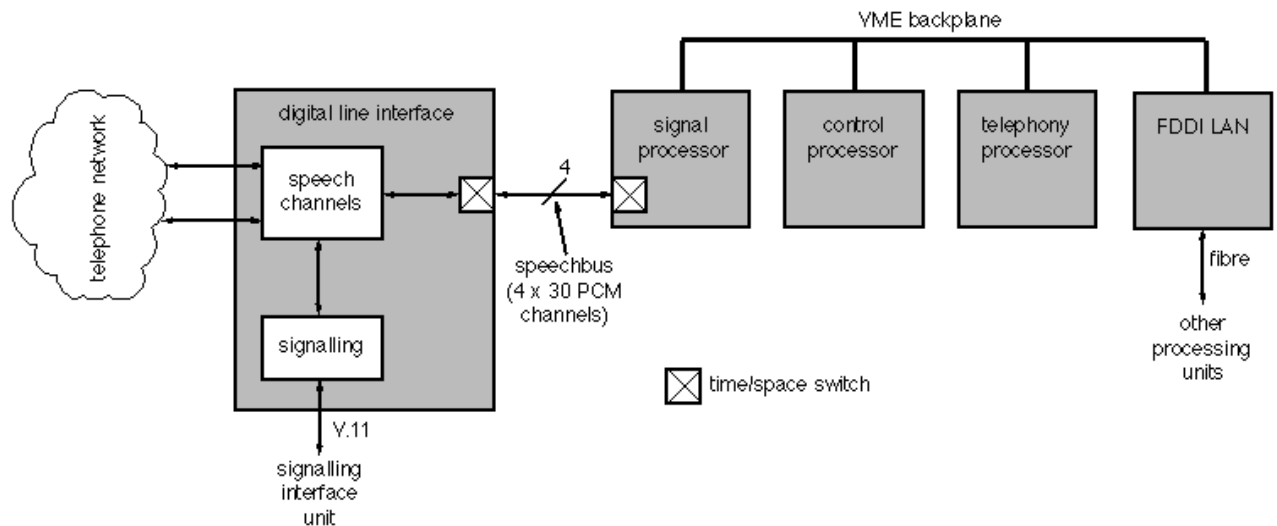


Fig 3 Speech processing unit functional block diagram.

system. As such, and like most large voice systems commercially available today, the SAP transmits control information and speech data on separate buses.

As the name suggests the SpeechBus is used for transmitting the speech data, leaving the VME bus for control information only. The SpeechBus has four 30-channel PCM highways that are used for transferring speech data between the digital line interface and the signal processors. Connection to the SpeechBus is via time/space switches located on the digital line interface and signal processor units. This provides the capability for the speech data to be switched internally within the SAP and creates the infrastructure required for dynamically allocating the speech resources (see later).

- **Telephony processor** — the telephony processor processes the C7 signalling NUP call-processing information from the telephony signalling unit and converts this information into a form that can be manipulated via the application programmers interface (API).
- **Control processor** — the control processor has two primary tasks: to manage the signal processing resources when requested by an application (via the API), and housekeeping. On receipt of an API request the control processor schedules the speech processing resources to undertake the task. This will include switching the speech data from the digital line interface to the appropriate signal processor units via the SpeechBus. As housekeeper, the control processor continually monitors the state of all modules to ensure that they are functioning correctly.

- **Signal processors** — they execute the various signal processing functions (e.g. speech recognition, speech recording, speech feedback) requested by the control processors.

4.4 Customer management units (data store)

These units are the mass storage systems which contain the customer profiles and customer speech data for different applications. In Fig 2 the customer management unit holds the customer data for the call minder call-completion service. The industry standard interfaces network file system (NFS) and small computer systems interface (SCSI) have been adopted for these units so that the appropriate technology can be utilised to meet the service requirements. In the case of the BT telephone network implementation, fixed magnetic disk technology has been used with each individual disk having a 'shadow' for resilience. The customer management units are shared between all the other units and the data is managed by using a relational database management system.

4.5 Management processor

This unit is the interface between the SAP system management and the BT network management system. It has three principal functions:

- to raise alarms to the network operations unit (NOU) via the BT network operations management system (NOMS) interface,

- to undertake system management commands requested by the network engineer in the NOU, e.g. the engineer may interrogate the status of a particular unit if an alarm is raised — these commands are transmitted from the engineer's system console in the NOU via the BT X.25 multi-protocol router network (MRPN)¹ with the statistical information, such as the number of calls handled, also provided upon request from the network engineer,
- to update customer information, e.g. add a new customer — this information would be transmitted from the customer service system (CSS) and collected by the operations and maintenance centre (OMC) software in a network administration computer centre (NACC) and transmitted on to the SAP via the MRPN.

All the SAP processor units can send and receive operational information, such as statistics, alarms, etc, to and from the management processor which then converts this information into the format required by BT's network management systems.

4.6 System back-up and data collector

As the name suggests this unit has two roles. As the system back-up unit it holds an up-to-date software image of the whole SAP system together with the applications software. Periodic back-ups are undertaken such that in the case of failure the system can be restored to the known state at the time of the last back-up. This back-up is stored on a magnetic tape cartridge.

As a data collector the unit can be programmed to monitor and record transactions undertaken on the SAP. For example, if the performance of the speech recogniser for a particular application needs to be assessed, then all the invocations of this application will be recorded by the data collector. This information is then stored on some form of bulk removable storage media, such as optical disks, and removed for analysis. The system back-up and data collector is usually located in the same place as the telephone exchange back-up tapes, e.g. local control point (LCP).

4.7 Local area network

All the above units communicate via an FDDI optical fibre LAN, although for smaller installations, an ethernet LAN is used.

4.8 Software overview

From a software perspective the SAP is a collection of independent software components each providing an integral part of the speech system. From a technical perspective these units have considerably different constraints placed upon them and the approach to the software design approach reflects this difference. In line with the design philosophy, either open and/or industry standards were adopted as appropriate. Although there are many software components, the key decision made for the software design was the choice of operating systems. Two operating systems were adopted:

- UNIX — for supporting the applications,
- VxWorks — a commercially available real time operating system that is very similar to UNIX in design and used for supporting the SAP control software.

A full description of the SAP software is not appropriate to this paper; however, to give a flavour of the software operation a single API command process is described.

Figure 4 shows a diagrammatic representation of the software processes involved in the execution of a typical API command.

The application processor is running the UNIX operating system whereas the remaining processors are running the VxWorks real time operating system.

For example, assume the application program issues the API command, 'Recognise_TouchTone_KeyPress'. The API passes this command to the application control process (that runs on the control processor) which in turn requests the resource for this function from the resource allocation process.

The resource allocation process reviews the current state of the signal processes (running on the signal processor unit) in the available pool and decides which of these should execute the TouchTone recognition — this is the heart of the dynamic resource allocation process adopted on the SAP. Several factors are taken into consideration when making the decision; for example, is a signal process resource available that has just executed a TouchTone recognition so that it can be reused and therefore avoid having to re-configure another resource?

Once the resource has been identified the resource allocation process informs the speech control process (that runs on the signal processor) of which process to run, on which resource to run the process, on which SpeechBus channel (not shown in Fig 4) to expect the incoming tone, and to which application process the recognised tone should be returned.

¹ Until recently, this X.25 network was known as the administration data packet network (ADPN).

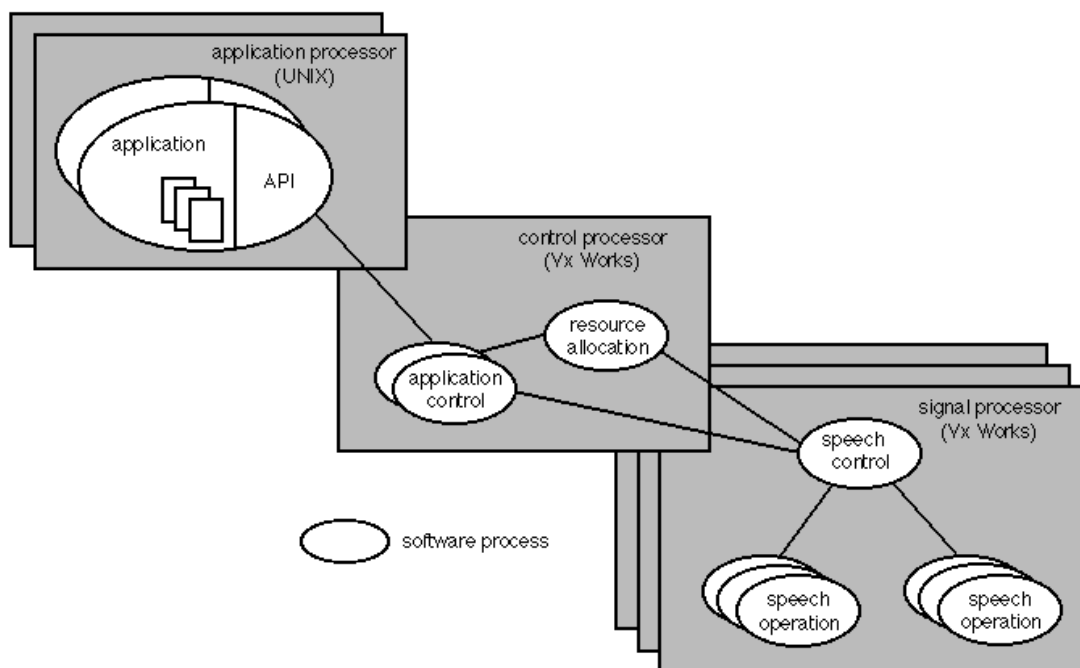


Fig 4 Typical SAP software operation.

The TouchTone recognition process is then performed (speech operation). Note that, in practice, many speech operations are being undertaken simultaneously. Once the process has been executed, the result (the recognised TouchTone digit) is passed to the speech control process and then on to the application control process. The speech control process also informs the resource allocation process that the signal processing resource used for the TouchTone recognition is now available.

Finally, the application control process returns the result of the TouchTone recognition to the application program via the API. The system operation is very complex owing to the large number of concurrent processes; however, it is this concurrent operation that provides the most optimal use of the available resources.

5. SAP as a service node

The service node architecture is covered in more depth in a separate paper [2]. Effectively, the only SAP component used in the service node architecture is the signal processor unit. All signalling and customer data handling are performed by other parts of the service node. The level of application control undertaken by the SAP is also determined by the service node.

6. SAP as an intelligent peripheral

It is planned that by 1996 the current SAP nodes in the BT telephone network will be modified to function as intelligent peripherals (IPs) as well as a 'stand-alone' speech sys-

tems. Figure 5 shows the modified SAP architecture that includes the IP functionality. The principal modifications lie in the telephony interface and how applications are managed.

- **Telephony interface** — the telephony interface is modified to run the C7 signalling connection control part (SCCP) as well as the message transfer part (MTP). The combinations of these software functions are known as the network service part (NSP). This upgrade gives the telephony interface compatibility with layer 3 of the open system interconnection (OSI) model.
- **IP service unit** — the IP does not require a CMU and instead an IP service unit is provided. This unit supports the C7 transaction capability application part (TCAP) and specialised resource function (SRF) to enable communication with a remote applications processor, known as a service control point (SCP). Service data, e.g. speech announcements, are stored as part of this unit.

For an IP application the applications manager (AM) (which runs on the application processor unit) will pass control of the application to the SCP which is usually located remotely from the SAP installation. The SAP communicates with the SCP using the intelligent network application part (INAP) SRF.

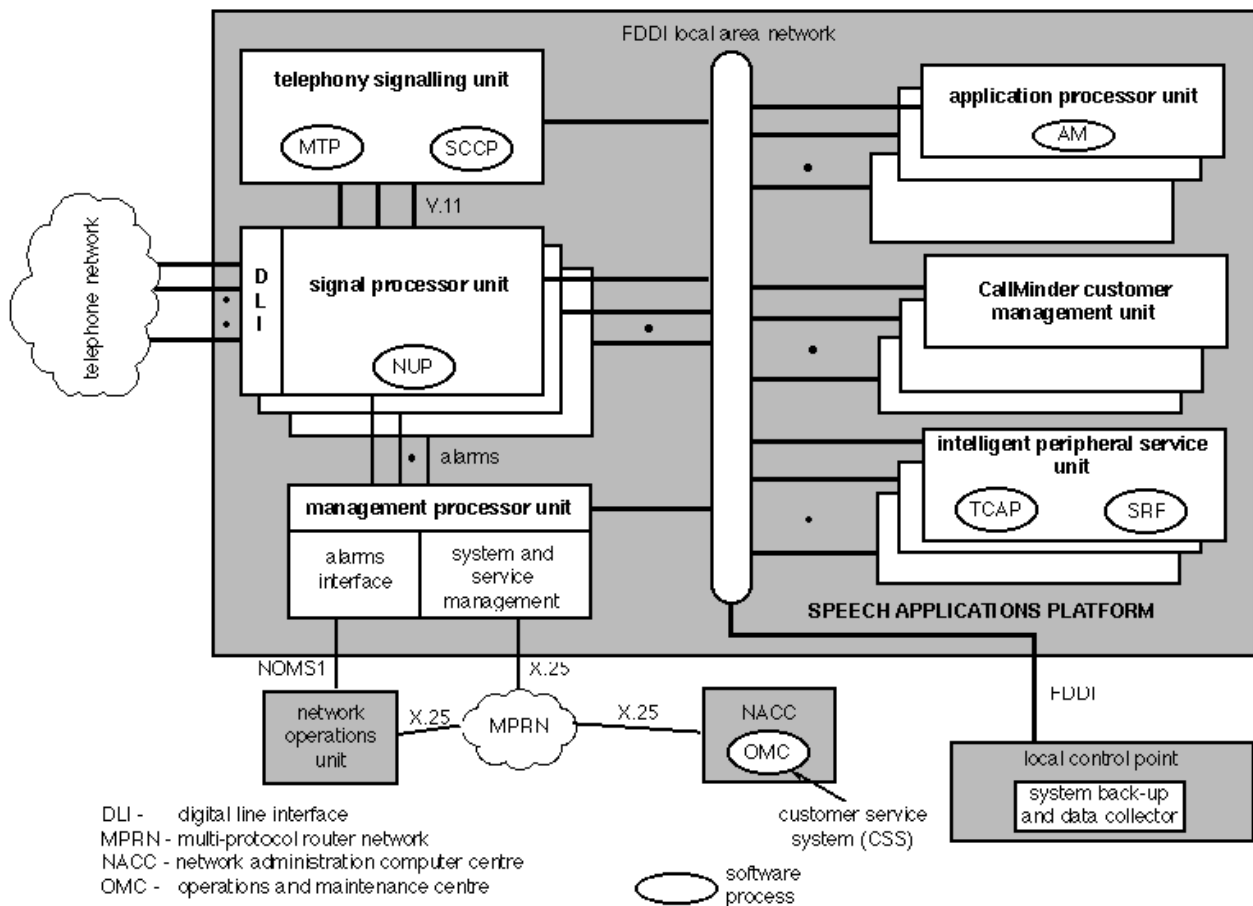


Fig 5 The BT SAP as an intelligent peripheral.

For a non-IP application the AM will interact with the local CMU to service the call and in doing so instruct a signal processing unit to execute the appropriate application.

7. SAP speech processing functionality

The following processing functionality is currently available on the SAPs integrated into the BT telephone network.

- TouchTone recognition — recognition of TouchTone digits transmitted from the customer/caller.
- Loop/disconnect transient recognition — recognition of loop/disconnect transients transmitted from the customer/caller.
- TouchTone generation — use of TouchTone digits to control remotely other items of telephony equipment, e.g. pagers.
- Speaker-independent speech recognition — the recognition of up to ≈ 30 words spoken by any person. This type of recognition would be used for the command words of a network service. The smallest vocabulary would consist of the words 'yes' and 'no' which would be used for very simple control of voice services such as call minder.
- Speaker-independent speech recognition — the recognition of up to ≈ 30 words spoken by a particular person. This type of recognition would be used when a customer has chosen specific words for a particular application, e.g. repertory dialling.
- Speech recording and playback at 64 kbit/s — speech data is recorded and played back at the standard telephony PCM rate. This facility is used when speech storage requirements are minimal.

- Speech recording and playback at 8 kbit/s — the speech is recorded at one eighth of the standard telephony PCM rate (64 kbit/s) which offers a major saving in data storage over other commercially available speech systems. The quality of the reproduced speech is at least as good (subjectively) as the speech feedback from telephone answering machines.
- Silence detection — this is used to detect when customers/callers have stopped speaking when recording speech.
- Call progress tone detection/generation — used for signalling on analogue telephony circuits.
- Modems — used in applications that require data transmission, e.g. remote meter reading.

It is planned to upgrade the SAP to include a number of new and advanced functions. These new functions will considerably expand the range of services which the SAP can support and further enhance its usefulness to BT.

8. SAP tools

Several tools have been developed to assist in using the SAP and are outside the scope of this paper. However, the visage application generation tool [3] should be mentioned in that it has significant benefits for the developer of network services. Visage is a PC-based tool with a graphical user interface that assists with design of voice services. The tool imposes a set of criteria upon the designer to ensure that the service developed meets those criteria defined in a style guide [4]. By using the tool it can be ensured that each service has the same 'look and feel'. Once the service has been designed it can be directly loaded on to the SAP application processor via the FDDI LAN. Therefore, the time taken to develop and test new services is dramatically reduced.

9. Conclusions

The speech system is an integral part of the intelligent network and can provide the sophisticated functionality necessary to control the advanced services offered by the intelligent network. However, the key to the successful take-up of network services will be the design of the customer interface. As the services become more and more sophisticated the challenge faced by human factors design engineers grows. It takes time for customers of such ser-

vices to learn to control the new features and to interact with services in a new or different way. As a result it may be that this learning time, and not technology, will determine the rate of take-up of these new services.

Acknowledgement

TouchTone, UNIX and VxWorks are registered trademarks.

Appendix

Definitions

For the purposes of this paper the following definitions have been adopted.

Customer — person who wishes to implement a network service (e.g. call diversion) or owns a network voice service (e.g. voice mail).

Caller — person who is met by a network or voice service when making a telephone call.

Telephone — the method by which a customer remotely controls access to a network service via the telephone network. Control is through the use of speech or call 'set-up' signals, i.e. loop-disconnect transients or dual tone multi-frequency (DTMF) tones — often referred to as TouchTone.

Network service — telephone management service; such services would include call diversion, message waiting, etc.

Voice service — automated customer voice services; such services would include call answering, voice messaging, etc.

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Kevin Rose joined BT Laboratories (BTL) as an apprentice in 1973. After graduating from Essex University with a BSc in computer systems and an MSc in telecommunications systems in 1983, he joined the newly established speech division in BTL. He has remained in the field of speech technology except for a short period in BT's switching systems business unit. In 1990 he became responsible for the development of the BT SAP and has managed its development from feasibility through to implementation into BT's telephone network. This has included managing the launch of the call minder pilot

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Pat Hughes graduated from Liverpool University with a BEng in electronic engineering in 1978 and went on to study for a PhD. Some six years later, having completed his doctorate in digital signal processing and spending three years as a lecturer, he joined BTL, working on speech analysis and synthesis. He became involved in the development of speech platforms, working initially with network-based message announcement systems. Later, he worked on the initial design of the SAP, and is currently co-ordinating its onward development.